The Grains of Structural Friction: Why AI Upskilling Is Failing the Power Law

The Premise

AI is often framed as a universal equaliser.

Give everyone access to the same tools.
Provide widespread training.
Raise the baseline—and outcomes should converge.

But this assumption is increasingly misaligned with reality.

Despite unprecedented access to AI tools and learning resources, outcomes are diverging—not converging. A small group compounds rapidly, a middle group makes incremental gains, and a long tail struggles to convert access into meaningful progress.

This is not just a productivity issue. It is a question of **structural divergence at scale**—with implications for economic mobility, workforce resilience, and social stability.

To understand why, we need to move beyond access and examine what truly governs outcomes.

The Hidden Variable: Structural Friction

The core constraint is not access. It is **structural friction**.

> **Structural friction is the total resistance an individual faces in converting access into capability—across cognitive, environmental, and behavioural dimensions.**

This includes:

- The mental models individuals bring
- The environments they operate within
- The habits, incentives, and identities they carry
- Their current capabilities

Two individuals may be equally exposed to AI. They may have access to the same tools, courses, and information.

Yet their ability to convert that access into meaningful outcomes can differ dramatically.

Why?

Because the effort required to move forward is not constant. It varies—sometimes subtly, sometimes exponentially.

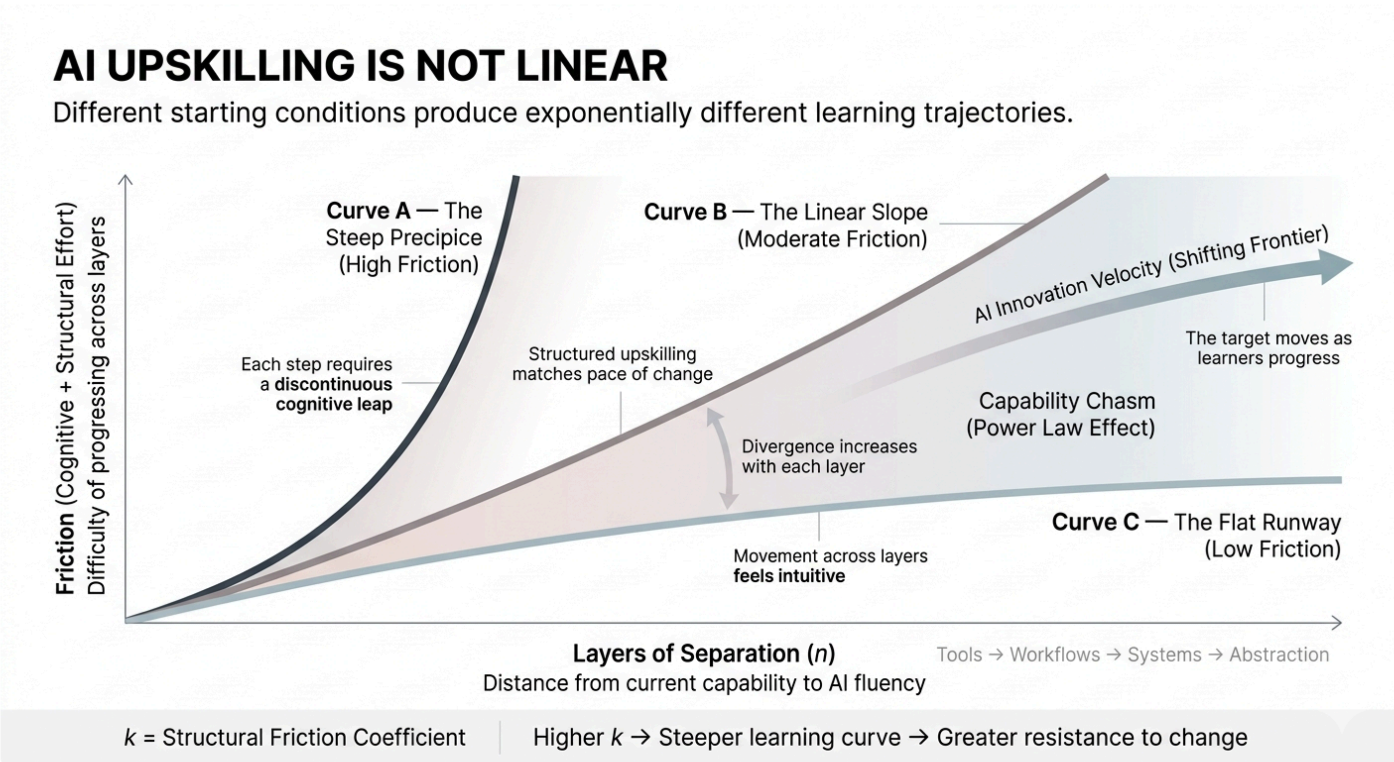
From Distance to Gradient

Most discussions of AI capability focus on **distance**:

- How far someone is from the technological frontier
- How much they need to learn

But distance is the wrong lens.

What matters is **gradient**—the effort required to close that distance.



- **Curve A: The Steep Precipice (Mature, Non-Digital SME)**
  For those shaped by decades of deterministic workflows, even the first shift requires a cognitive leap. There is no gradual incline—only discontinuity.

- **Curve B: The Linear Slope (Mid-Career PMET)**
  For professionals in structured, digitized environments, the climb is steady. Corporate training scaffolds the journey. Progress is real—but bounded by context.

- **Curve C: The Flat Runway (AI-Native)**
  For those with computational or design-native backgrounds, moving across AI layers feels intuitive. The friction is minimal. Each new tool maps naturally onto existing mental models.

Two individuals may be equally far from the frontier.
One faces a manageable slope.
The other faces a near-vertical wall.

The difference is not visible in access.
It is embedded in structure.

When Outcomes Follow a Power Law

Once we account for gradient, a familiar pattern emerges: **power-law outcomes**.

In such systems:

- A small group captures disproportionate value
- A middle group sees incremental improvement
- A long tail struggles to translate effort into results

This is where most AI upskilling narratives break down.

They assume that uniform inputs—courses, subsidies, tool access—will produce broad-based uplift.

But power-law systems do not respond to uniform inputs. They amplify initial differences.

The same intervention that accelerates the top 10% may leave the bottom 50% largely unchanged.

The Compounding Effect of Speed

Even this framing is incomplete.

If structural friction were the only constraint, time and persistence might still close the gap.

But AI does not stand still.

What is current today becomes obsolete in months. Interfaces evolve, paradigms shift, and expectations reset continuously.

For those on low-friction curves, speed compounds advantage. They adapt in real time and extend their lead.

For those on high-friction curves, speed is destabilising. Just as a foothold is gained, the ground shifts again.

The result is not just a widening gap—it is a **moving target** with accelerating divergence.

From Gap to Moving Staircase

When adaptation constraints are non-linear, the dynamics change fundamentally.

- Those facing lower friction move faster
- The technological frontier advances simultaneously
- The distance to relevance does not shrink—it shifts

This creates what can be described as a **moving staircase effect**:

Some are climbing. Some are standing still. But the staircase itself is rising.

The gap does not merely widen. It accelerates.

Upskilling is no longer about catching up. It is about chasing a horizon that keeps receding—and questioning whether catching up still matters by the time one gets there.

Over time, if this dynamic persists, we may see the emergence of disengagement behaviours—sub-cultures of “lying flat(躺平)” to withdrawal(ヒキコモリ)—bringing broader social and policy implications.

Why Most AI Efforts Converge to Average

Without a clear understanding of these dynamics, individuals and organisations respond predictably:

They try to move faster.
They adopt tools aggressively.
They experiment continuously.

But without proper diagnosis, this often results in:

- Poor problem definition → faster wrong solutions
- Shallow understanding → more polished but average outputs
- Misaligned goals → efficient irrelevance

AI does not fix flawed thinking. It scales it.

The Failure of One-Size-Fits-All Solutions

Most interventions remain structurally uniform:

- Standardised courses
- Broad-based subsidies
- Generic “AI literacy” programmes

These approaches assume that access is the primary constraint.

It is not.

The real constraint is **misdiagnosis**.

In a power-law system, the average solution serves no one:

- Too shallow for those who need transformation
- Too slow for those who are already accelerating
- Misaligned for those facing structural discontinuity

Reframing the Problem: From Content to Cognition

If the issue is structural, the solution cannot be purely curricular.

We must shift from **what we teach** to **how individuals think and operate**.

This requires a more precise diagnostic lens:

- What mental models does the learner already possess?
- How adaptable is their working environment?
- How much unlearning/relearning is required before new capabilities can take hold?

For some, the barrier is technical knowledge. For others, it is identity, habit, or worldview. For many, it is the interaction of all three factors.

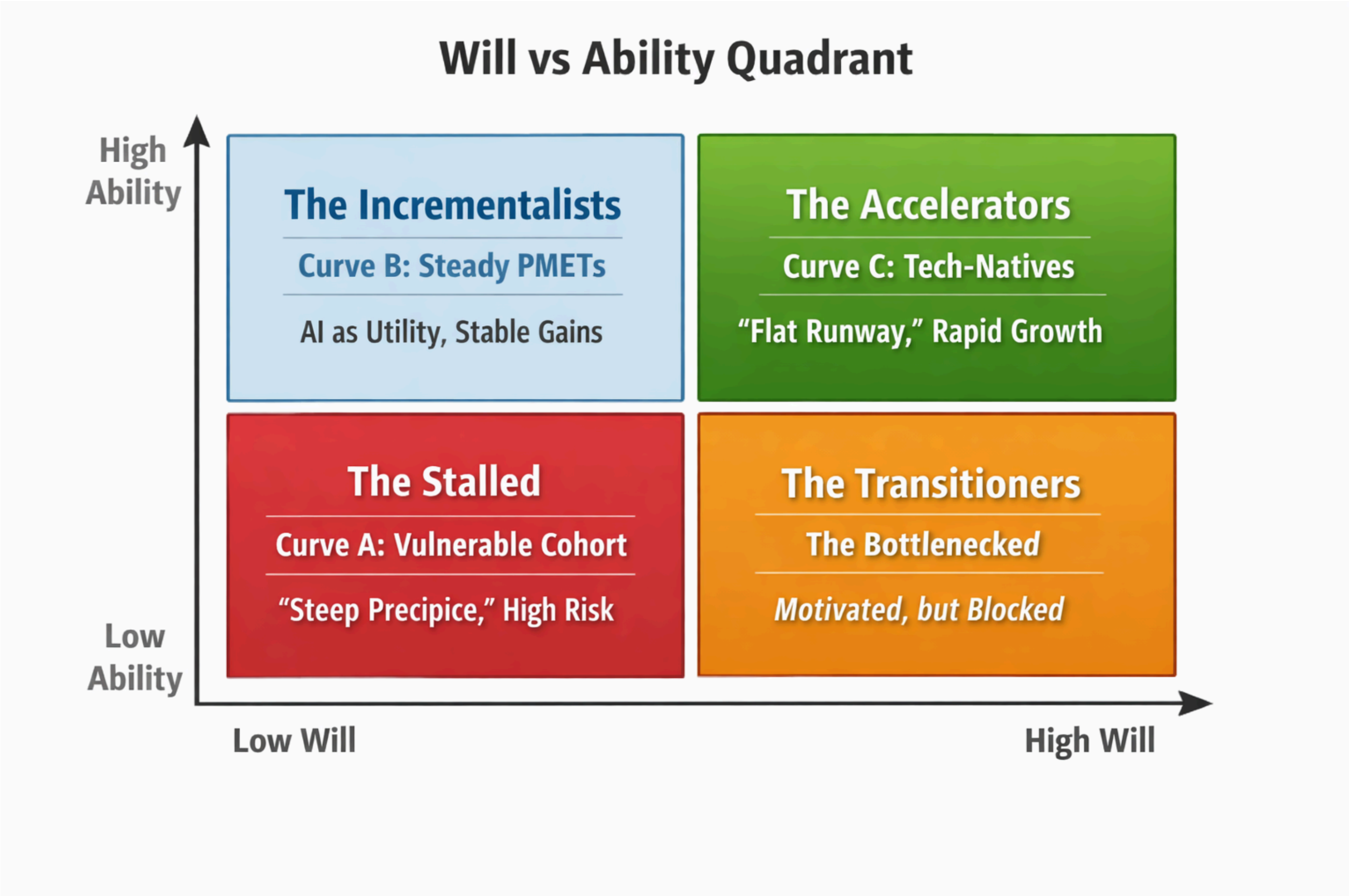
Until these factors are surfaced, upskilling efforts will remain blunt instruments.

Designing for Divergence, Not Uniformity

A more effective approach begins with divergence as the baseline.

It must account not only for capability, but also for intent.

An **ability–motivation matrix** provides a practical way to operationalise this:



Group	Description	What They Need
The Natives	High ability, high motivation	Remove constraints, maximise leverage, enable rapid experimentation
The Integrators	Moderate ability, high motivation	Redesign workflows, embed AI into existing systems, reduce integration friction
The Constrained Adopters	Low ability, high motivation	Simplify tools, reduce cognitive load, provide guided pathways
The At-Risk	Low ability, low motivation	Reframe relevance, rebuild motivation, address identity and behavioural barriers

The earlier gradient model explains **why outcomes diverge**. This matrix provides a way to **act on that divergence**.

Implications: Beyond Upskilling

For policymakers

Upskilling remains necessary—but it is insufficient on its own.

Equal access to training does not produce equal outcomes when structural friction varies. Interventions must account for starting conditions, not just curriculum delivery.

For founders and builders

The next wave of value lies not in building more AI tools—but in **reducing friction**.

Opportunities lie in:

- Abstraction
- Simplification
- Humanising interfaces

Not everyone can move faster. But systems can be designed to require less movement.

For economies

In highly connected and urbanised societies, uneven acceleration introduces structural risk.

It erodes the stability of the traditional bell curve and creates sharper divides.

At the same time, maintaining competitiveness requires continuous improvement of the national average.

How these gradients are managed will shape both economic performance and social cohesion.

Singapore as a Baseline

Singapore's coordinated and well-funded approach provides a useful baseline.

It does not invalidate the gradient framework—it reinforces it.

If structural friction emerges even in a high-trust, policy-aligned environment, it suggests that similar dynamics will manifest more strongly elsewhere—particularly in regions with fewer institutional buffers and steeper cognitive barriers.

The Bottom Line

AI has the potential to be either a universal equaliser or a powerful divider.

Access is necessary—but far from sufficient.

In a power-law system, equality of input does not produce equality of outcome. Structural friction determines who compounds—and who stalls.

We can understand this through a **Power-Law Constraint Model**:

> Outcomes are shaped by the interaction between structural friction, compounding dynamics, and a continuously shifting technological frontier.

The divide will not be closed by giving everyone the same curriculum.

The risk is not that AI leaves people behind.
It is that it accelerates unevenly—and we mistake that acceleration for progress.

Until we learn to see the gradients, we will continue to optimise for motion rather than meaningful improvement.

Over time, these gradients do not disappear.
They accumulate—settling into the grains of structural inequality.

One-Line Summary

> In a power-law system, progress is determined not by how far you are—but by how hard it is for you to move, and how fast the world is moving while you try.
